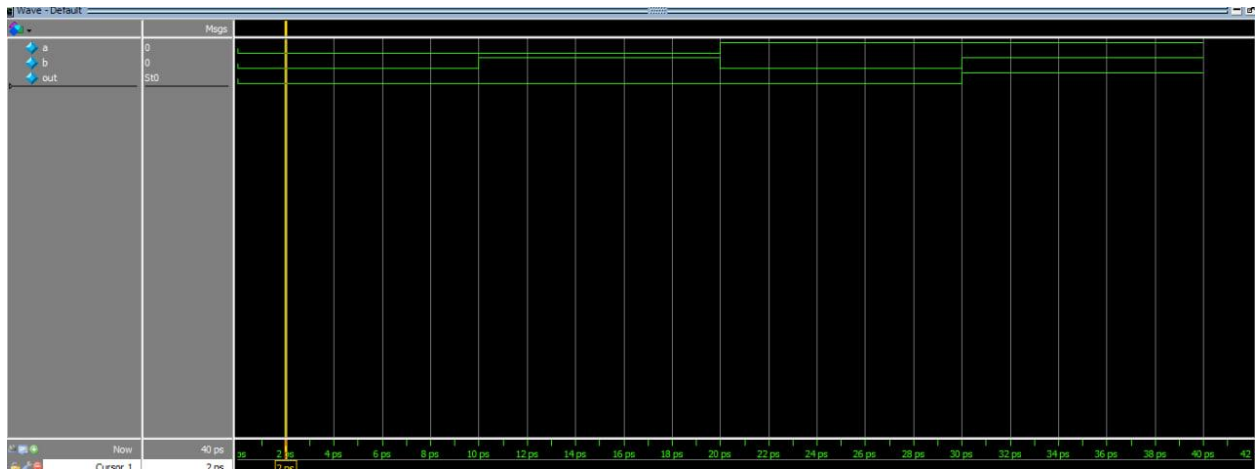
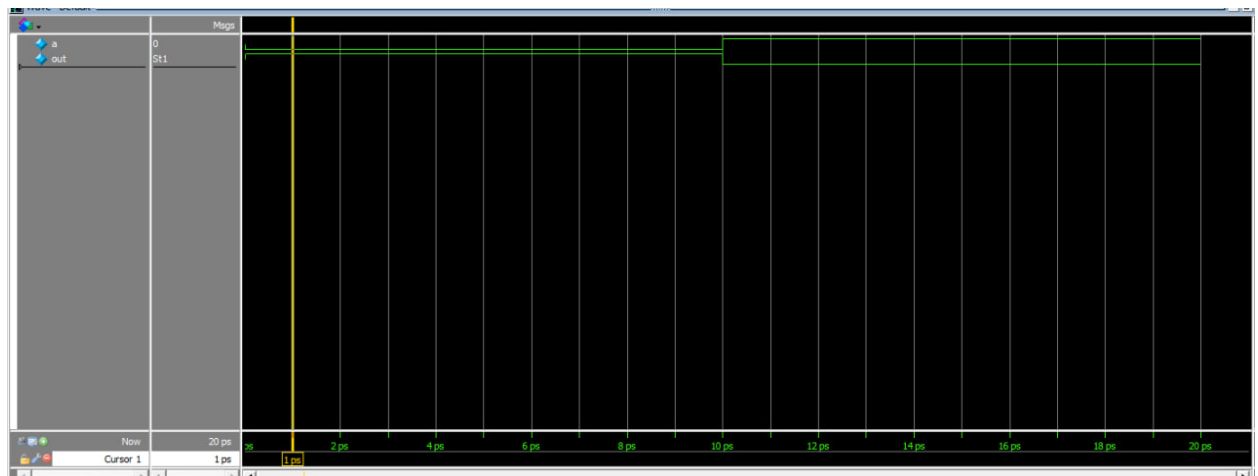


Waveforms:

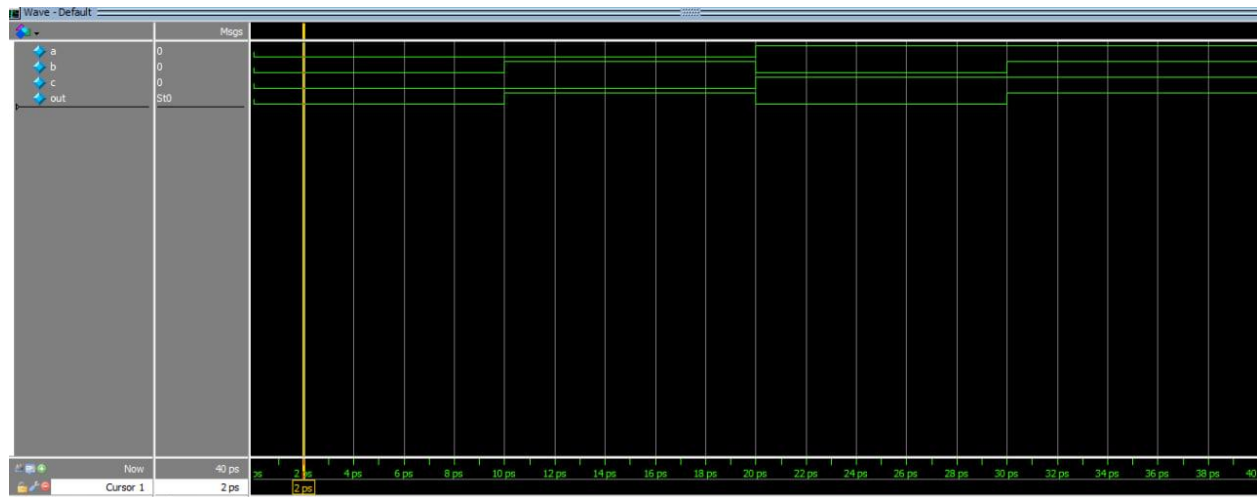
And1bit:



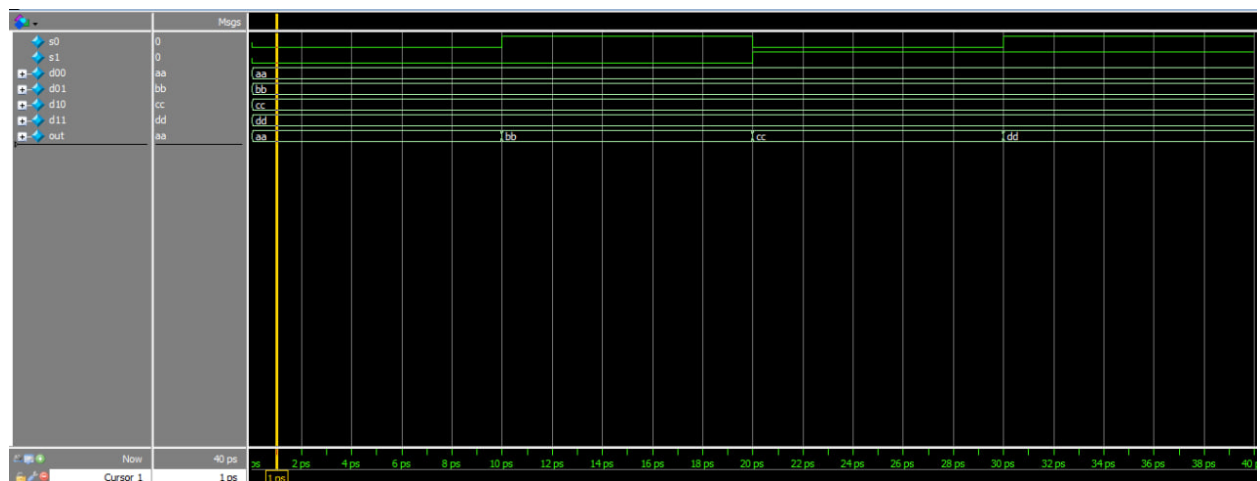
Not1bit:



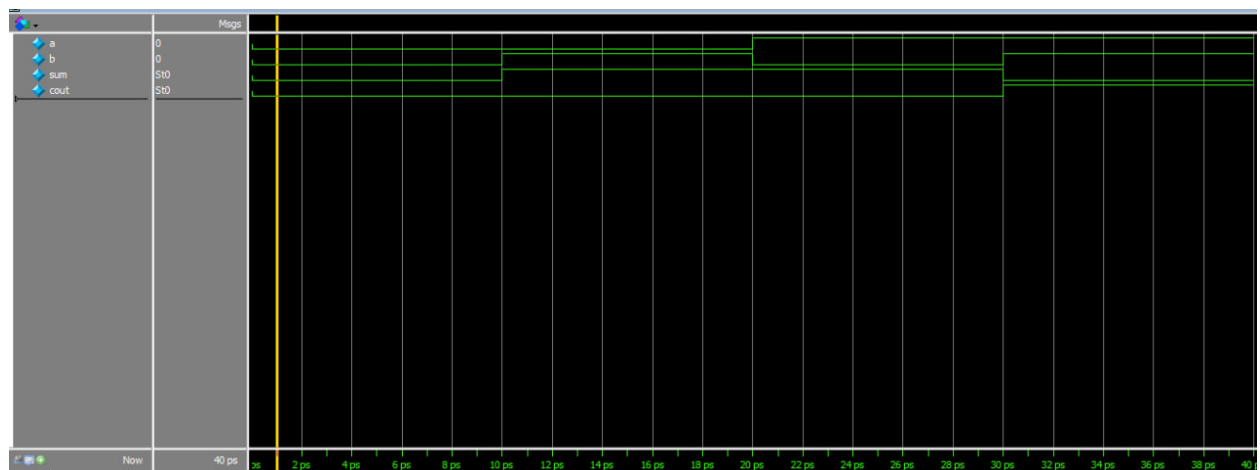
Xor3:



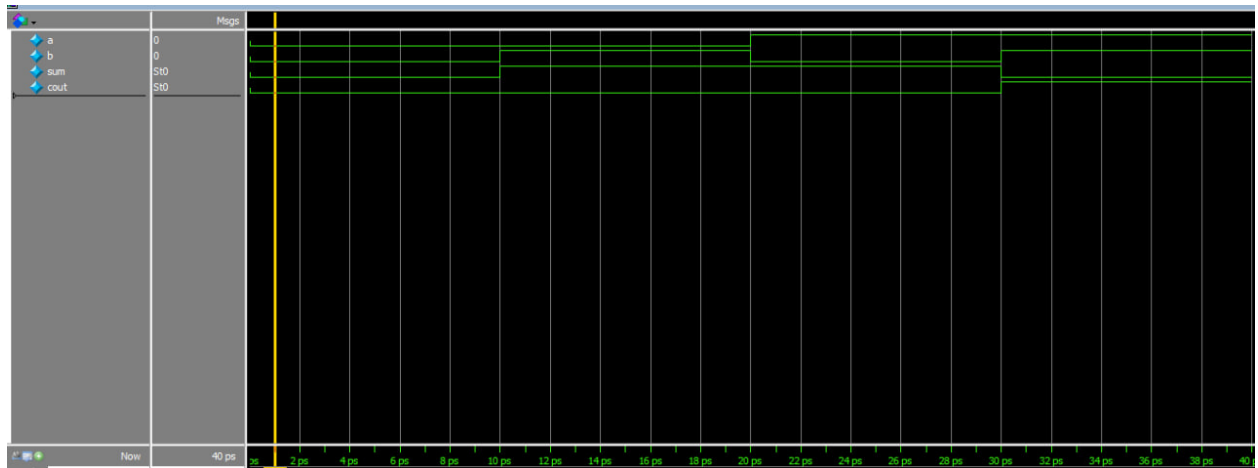
Mux4to1:



Half adder:



Full adder:



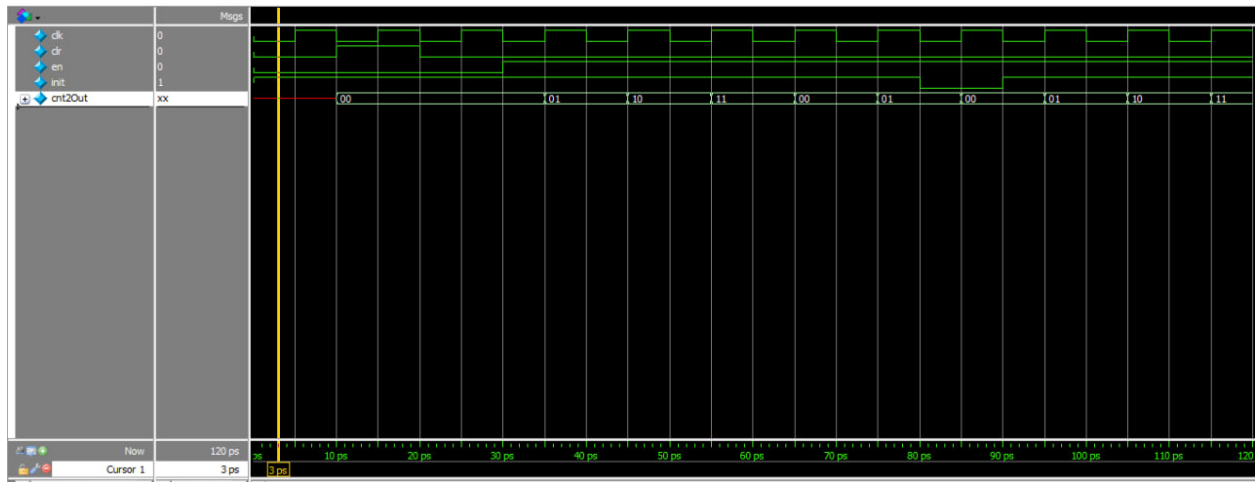
Normal register:



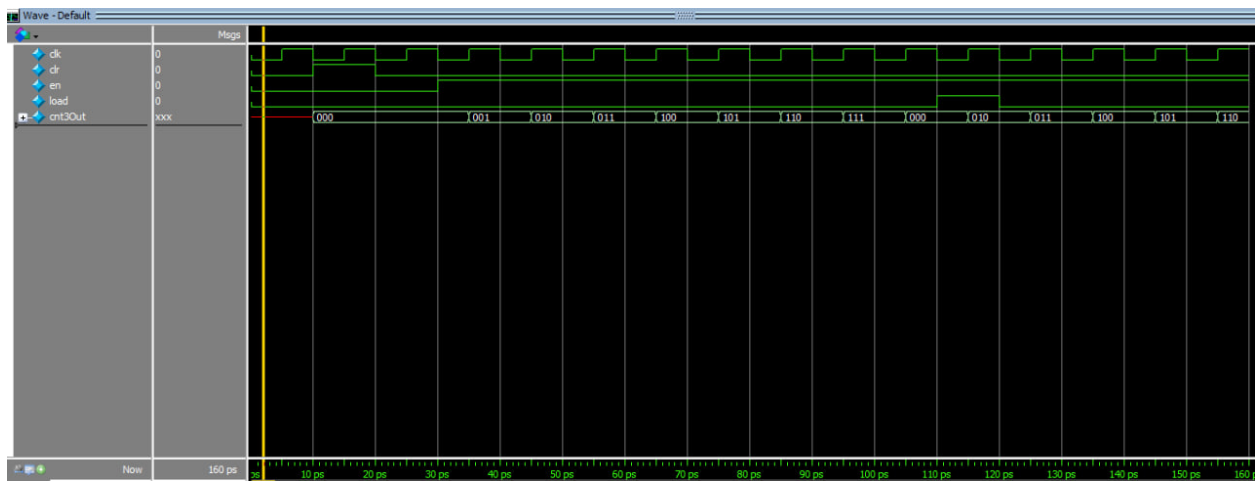
Load register:



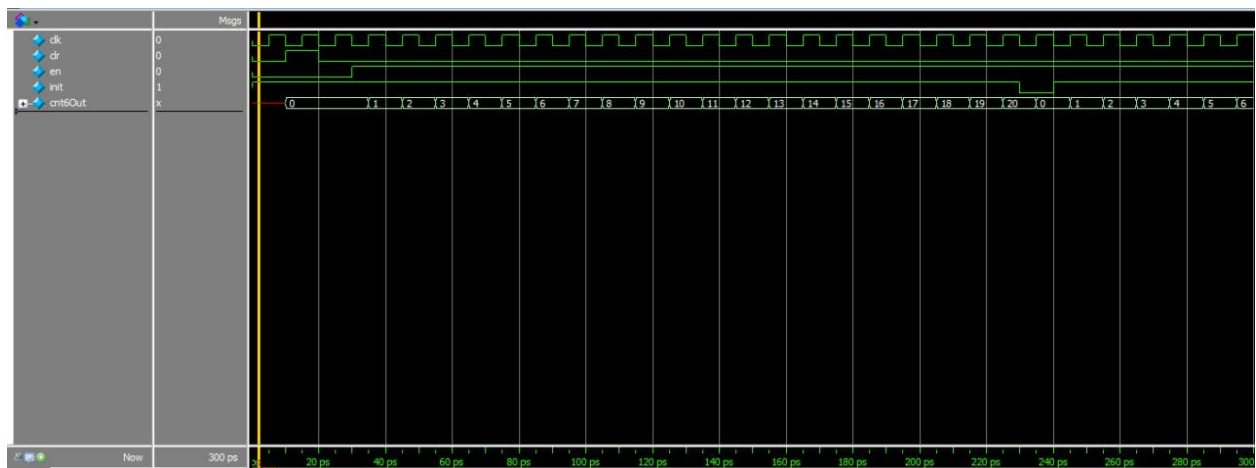
Counter2bit:



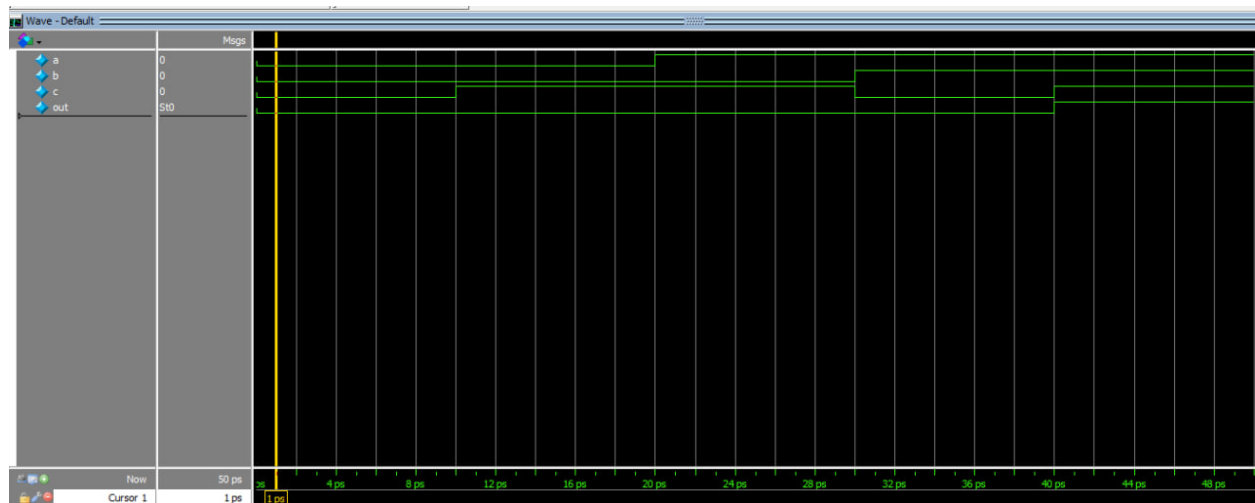
Counter3bit:



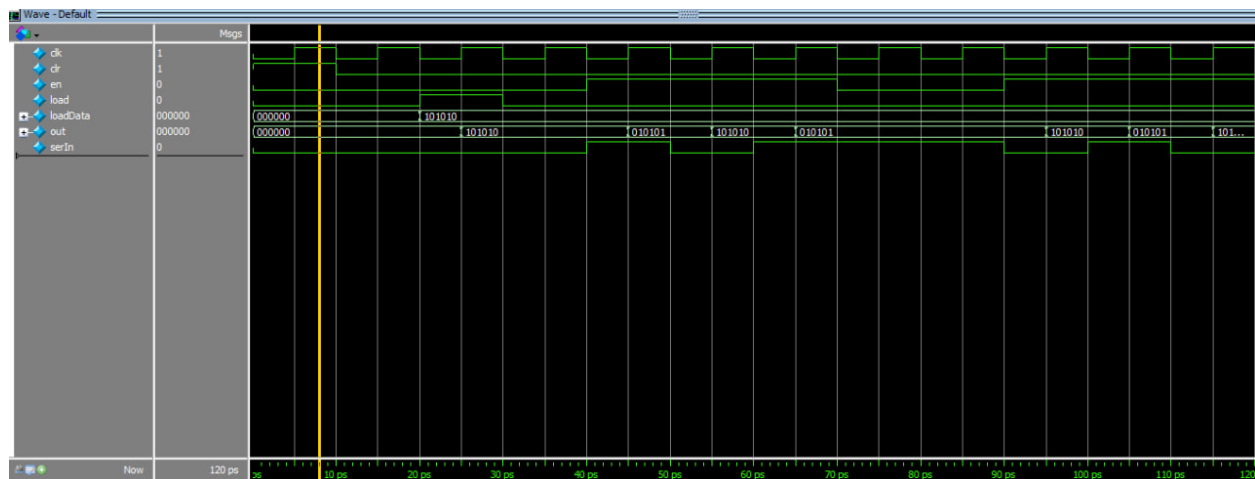
Counter6bit:



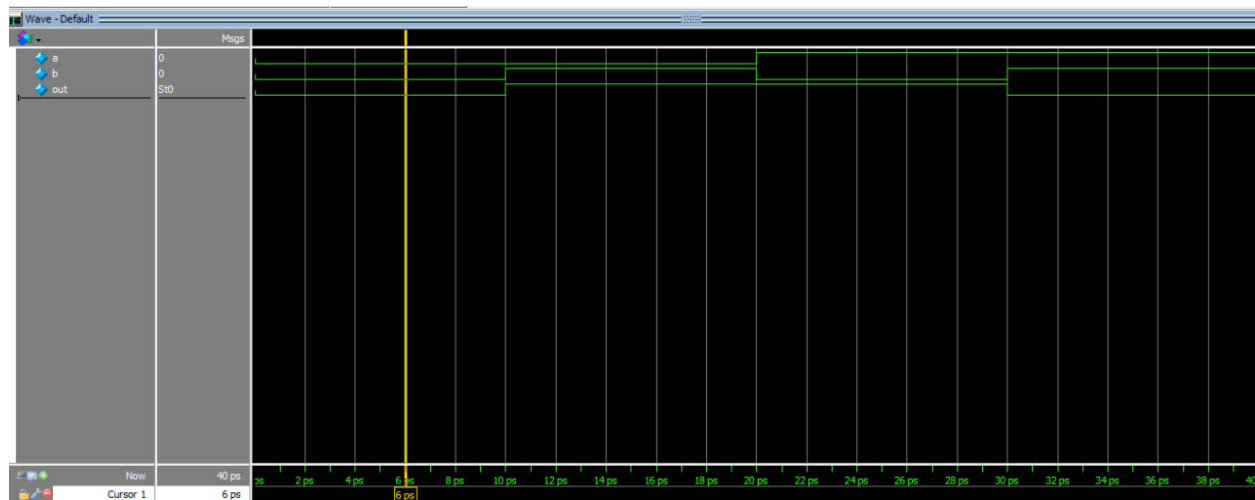
And3:



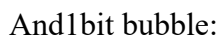
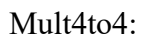
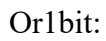
Shift register 6bit:



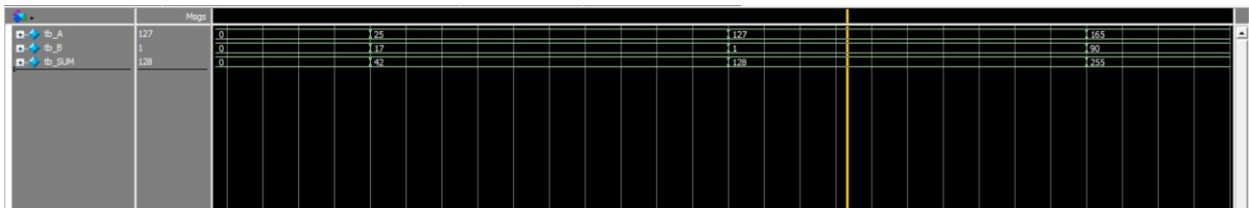
Xor2:



And4:



Adder:

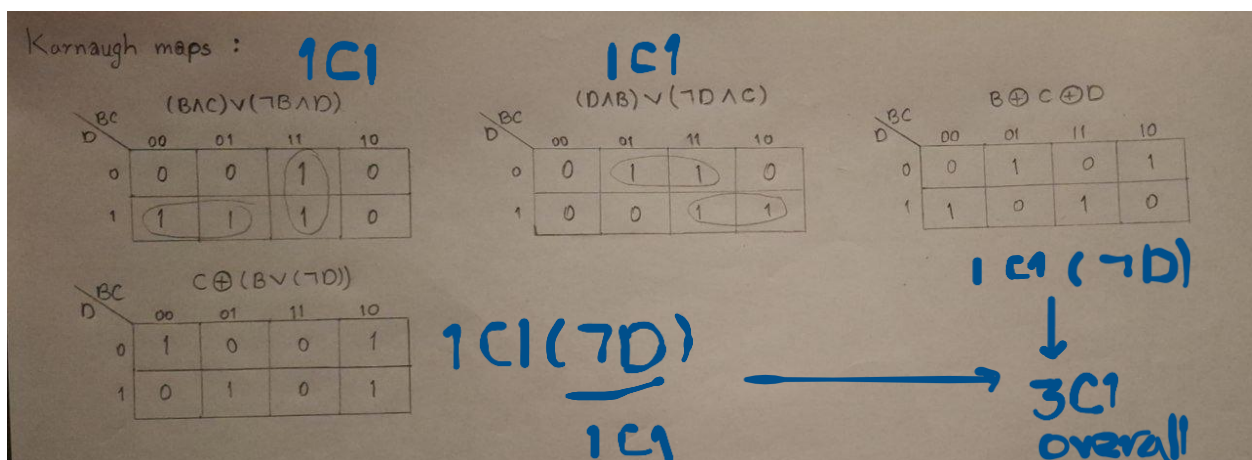


Random generator:



Hash generator:

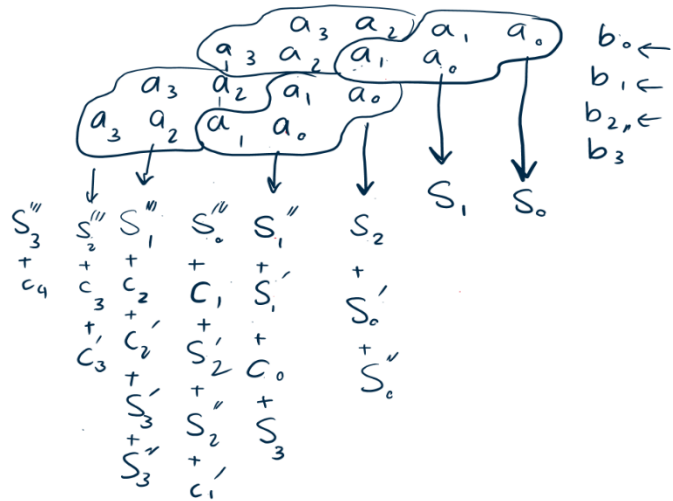
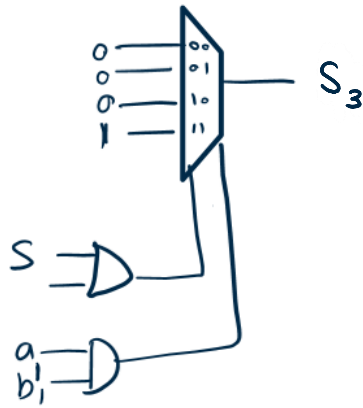
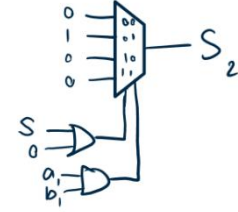
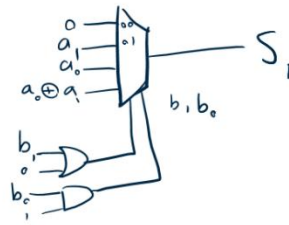
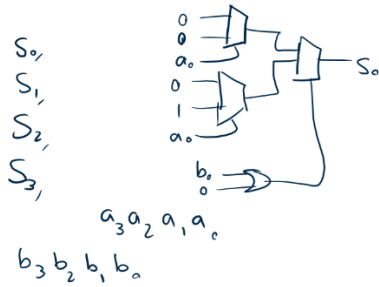
Karnaugh maps:



Other logic diagrams:

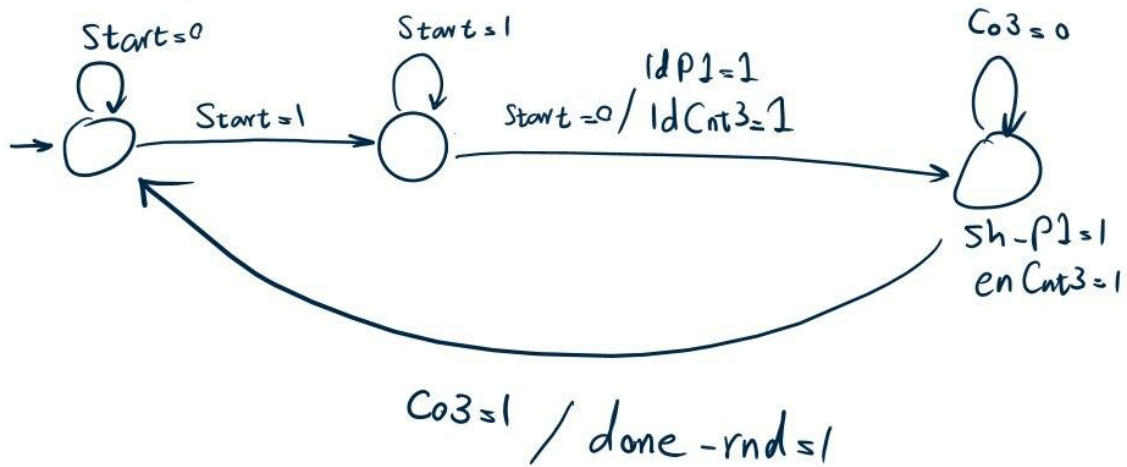
Multiplier:

2 to 2 Multiplier

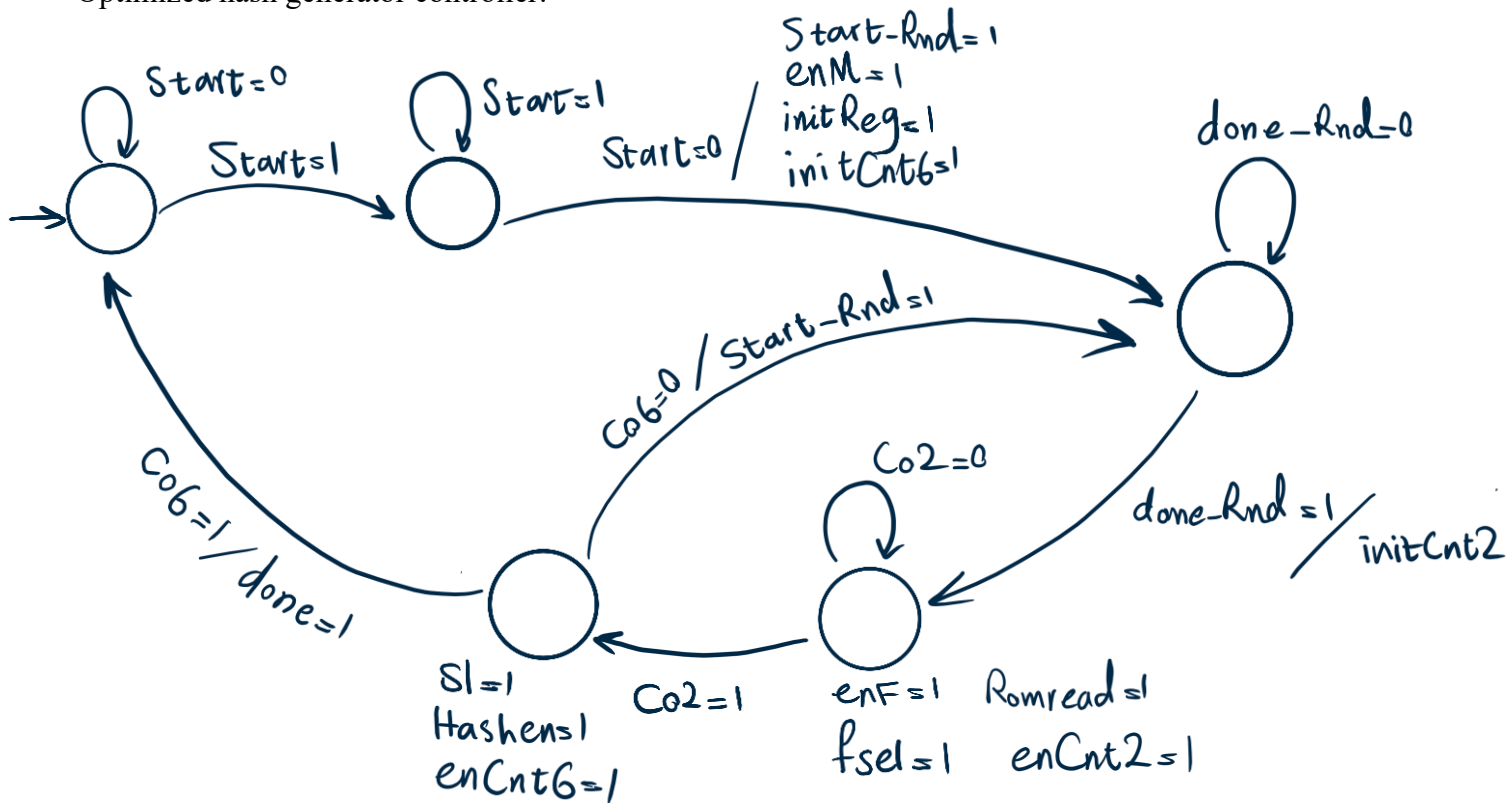


Optimized random generator controller:

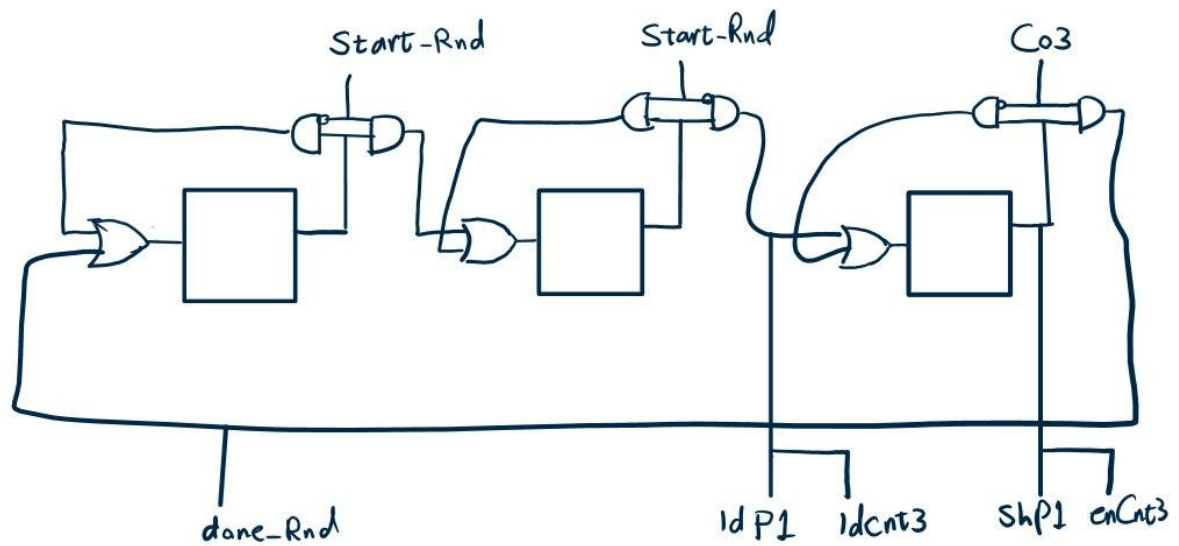
part-1



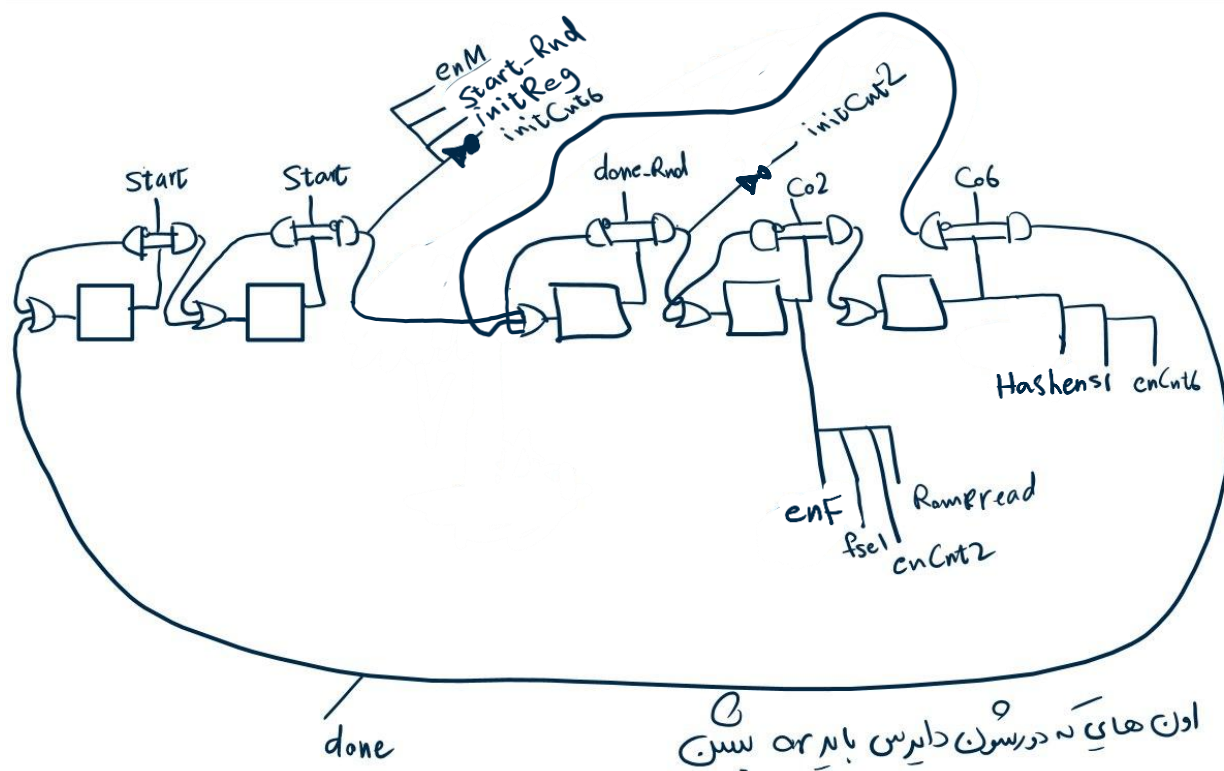
Optimized hash generator controller:



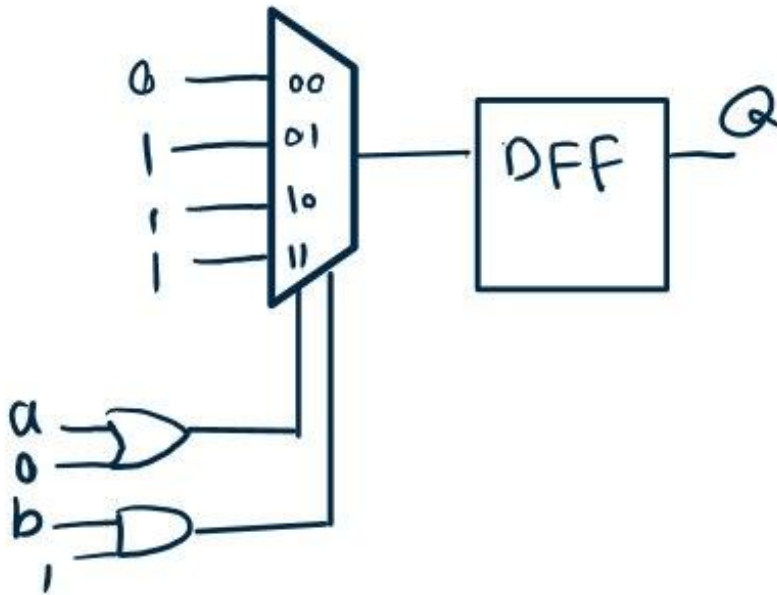
Structural design for random generator controller:



Structural design for hash generator controller:

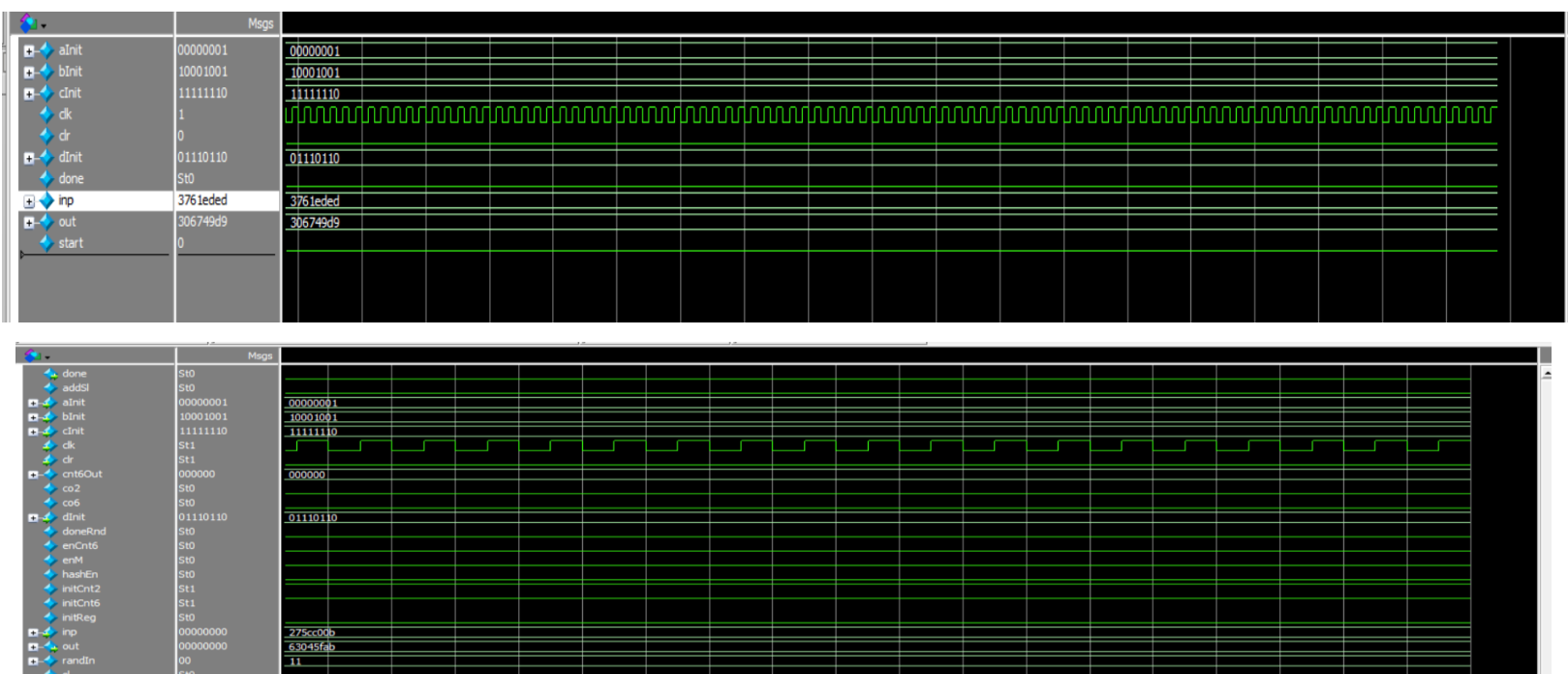


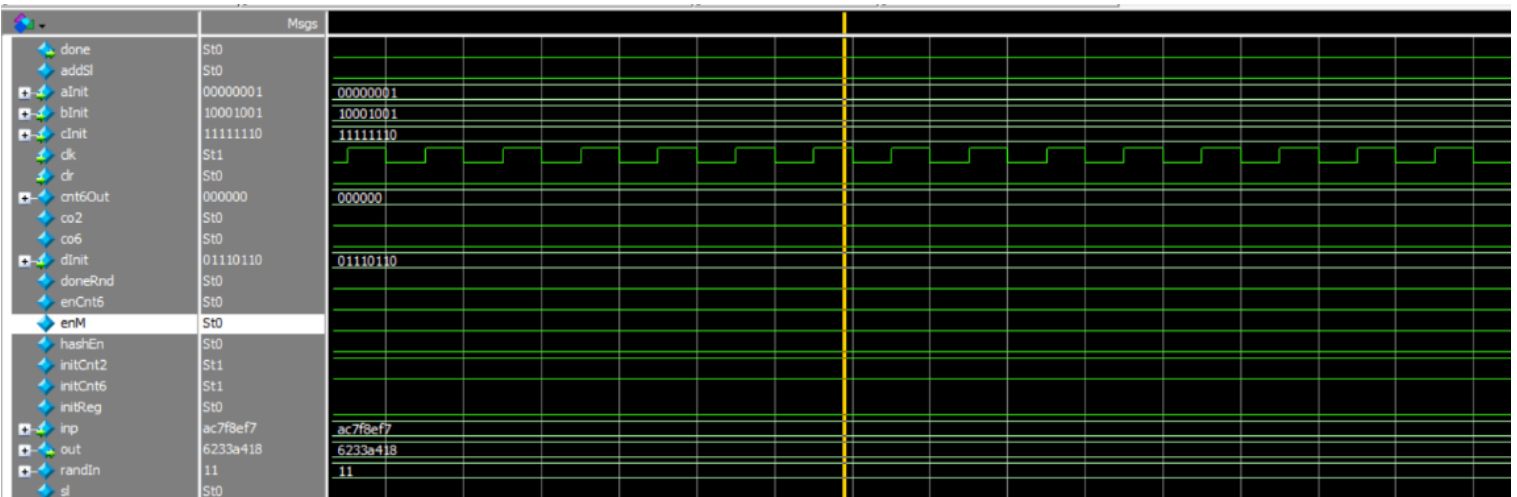
Where each logic block will look like this for each of the bits for one-hot states:



Also there should be a not gate for the first bit of the controller DFF because when we try to reset the state it should be turned 1 not 0.

Also you can see verification of the circuit as follow:





And this is number of our used gates:

numbers.txt

1 2982